PART 1

Center the Data

points: $(-2,-1), (-1,0), (0,1), (1,2), (2,3)$

mean: $\mu = \left(\frac{-2-1+1+2}{5}, \frac{-1+1+2+3}{5} \right) = (0,1)$

points − mean: $(-2,-2), (-1,-1), (0,0), (1,1), (2,2)$

Find 1st Principal Component

all centered points lie on $x_2 = x_1$, so this dir. gives no orthogonal reconstruction error. Normalizing $(1,1)$ gives the first principal component.

$w_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\quad \|w_1\|_2 = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1$

Reconstruction Error for (3,3)

centered point: $x_c = x - \mu = (3,2)$

coordinate along 1st PC: $z = w_1^T x_c = \frac{3+2}{\sqrt{2}} = \frac{5}{\sqrt{2}}$

reconstruct + mean: $\hat{x} = \mu + z w_1 = (0,1) + \frac{5}{\sqrt{2}} \frac{(1,1)}{\sqrt{2}} = \left(\frac{5}{2}, \frac{7}{2} \right)$

$\therefore$ L2 reconstruction error: $\|x - \hat{x}\|_2 = \sqrt{\left(3 - \frac{5}{2}\right)^2 + \left(3 - \frac{7}{2}\right)^2} = \frac{1}{\sqrt{2}} \approx 0.7071$

Reconstruction Error for (3,4)

$x_c = \text{point} - \mu$
$= (3,4) - (0,1) = (3,3)$

project: $z = w_1^T x_c = \frac{3+3}{\sqrt{2}} = \frac{6}{\sqrt{2}}$

reconstruct: $\hat{x} = (0,1) + \frac{6}{\sqrt{2}} \frac{(1,1)}{\sqrt{2}} = (0,1) + (3,3) = (3,4)$ $\quad \therefore, \|x - \hat{x}\|_2 = 0$

This point lies on the PCA line $x_2 = x_1 + 1$